



# SS26 MTM-Setup – User manual

Welcome to the user manual for the MTM-Setup. This document is designed to help you understand and efficiently operate your stepper motor using a series of predefined commands. Each command has specific parameters and usage guidelines to ensure precise control and accurate responses from the system.

## Setup System Requirements

- Control unit
- Stepper motor
- Force sensor unit
- External power supply (24V / 1.0A)
- Terminal program for communication

## Setup instruction

- Connect the motor (M) to the control unit via the MX1 and MX2 ports.
- Connect the amplifier box (LC) with the force sensor to the control unit at the designated 'F Sensor' port.
- Attach the power supply to the port labeled 'Power 24V'.
- Switch on the power supply at the rear, adjust the power supply to 24V and set the current to 1.0A.
- Activate the voltage output to the control box using the front 'Output' button.
- Ensure the red button is disengaged to enable motor operation.

To identify the correct port using Device Manager:

- Access Device Manager by pressing Win + X and selecting 'Device Manager' (Geräte-Manager).
- Expand the 'Ports (COM & LPT)' section.
- Connect the control unit via USB serial adapter.
- Observe the new COM port that appears.
- Note the assigned COM port number (e.g., COM8) for use in your terminal program.

## Starting the Terminal Program

- Open YAT and the terminal '...'.
- Verify if the terminal is connected to the correct port under Terminal Settings.

->Terminal -> Setting

Port: COM8 - Serielles USB-Gerät - (in use by this terminal) 

Bits per Second: 115200

Data Bits: 8

Parity: None

Stop Bits: 1

- If the right port is not listed, click the refresh symbol next to the port selection dropdown.
- Click on the green symbol "Open/Start Terminal" to establish the connection to the terminal.



- Press the reset button on the control unit

The system response should look like this:

```
#####<CR>
#####<CR>
MTM_Setup V1.00<CR>
#####<CR>
I2C_EEPROM_VERSION: 1.8.2<CR>
EEPROM is connected.<CR>
<CR>
.The calibration value from the EEPROM is: 1104.23<CR>
.HX711 measured conversion time ms: 10.67<CR>
.HX711 measured sampling rate Hz: 93.75<CR>
.HX711 measured settlingtime ms: 172<CR>
.Load-Cell startup is complete<CR>
Motor <DRIVER DM542T> initialization done!<CR>
#END SETUP<CR>
#####<CR>
UNIT IS MILLIMETER<CR>
#####<CR>
```

After this, you can enter commands and receive output from the device.

**The first command should be the ,Init'- command (see command table). Alternatively, instead of entering the init command, you can press and hold the Init button for 3 seconds.**

If syntax errors occur when entering commands, error messages in the form of "ERR -1/2/3/4/5/6:<CR>" will be displayed.

**Note: The motor works exclusively in steps, with each step corresponding to 0.0635 mm. This means that movement commands that specify positions and speeds in millimeters do not result in exact millimeter movements, but in steps of 0.0635 mm. Users should take this into account when configuring precise movements or measurements.**



## Predefined Commands

|             |                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Init</b> | Command                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|             | Description:                | Command for the initialisation run, finds position zero.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|             | Syntax:                     | „Init“ ( <i>No arguments required</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|             | System response:            | <code>.button_S3_pressed</code><br><code>.button_S3_released</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>L</b>    | Command                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|             | Description:                | Prints current position.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|             | Syntax:                     | „L“ ( <i>No arguments required</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|             | System response:            | <code>.Current location of the motor [ mm]: 0.000000&lt;CR&gt;</code><br><code>.Current location of the motor [steps]: 0&lt;CR&gt;</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>S 0</b>  | Command                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|             | Description:                | Retrieves system parameters including unit of measurement (mm), maximum acceleration values in both mm/s <sup>2</sup> and steps/s <sup>2</sup> , maximum speed values in mm/s and steps/s, and minimum and maximum position values in both mm and steps.                                                                                                                                                                                                                                                                                                                                                         |
|             | Syntax:                     | „S 0“                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|             | Example of system response: | <code>.Unit: mm&lt;CR&gt;</code><br><code>&lt;CR&gt;</code><br><code>.Max acceleration value [mm/s^2]: 635000.0&lt;CR&gt;</code><br><code>.Max acceleration value [steps/s^2]: 1000000.0&lt;CR&gt;</code><br><code>&lt;CR&gt;</code><br><code>.Max speed value [mm/s]: 444.50&lt;CR&gt;</code><br><code>.Max speed value [steps/s]: 7000.00&lt;CR&gt;</code><br><code>&lt;CR&gt;</code><br><code>.Min position [mm]: 0.000000&lt;CR&gt;</code><br><code>.Max position [mm]: 120.014999&lt;CR&gt;</code><br><code>.Min position [steps]: 0&lt;CR&gt;</code><br><code>.Max position [steps]: 1890&lt;CR&gt;</code> |
| <b>S 1</b>  | Command                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|             | Description:                | Prints the calibration value.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|             | Syntax:                     | „S 1“                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|             | System response:            | <code>Calibration value is: 1104.23&lt;CR&gt;</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>S 2</b>  | Command                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|             | Description:                | Prints the debug visibility                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|             | Syntax:                     | „S 2“                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|             | System response:            | <code>Debug visibility: 0&lt;CR&gt;</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>S 3</b>  | Command                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|             | Description:                | Prints the bar graph parameters. See also command <b>BarGraphValue</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|             | Syntax:                     | „S 3“                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|             | System response:            | <code>.minValue[Newton] : -10.0&lt;CR&gt;</code><br><code>.quietValue[Newton]: 0.0&lt;CR&gt;</code><br><code>.maxValue[Newton] : 10.0&lt;CR&gt;</code><br><code>.barDisplayWidth : 101&lt;CR&gt;</code>                                                                                                                                                                                                                                                                                                                                                                                                          |



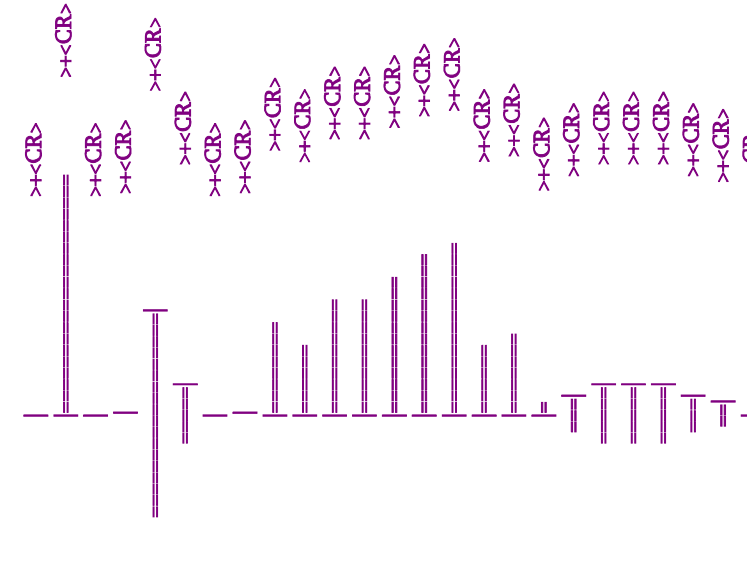
|                      |                                      |                                                                                                                                                                                                                          |
|----------------------|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>BarGraphValue</b> | Command Description:                 |                                                                                                                                                                                                                          |
|                      | Syntax:                              | „BarGraphValue <i>Parameter1 Parameter2 Parameter3</i> “                                                                                                                                                                 |
|                      | Parameter units                      | Parameter1 minValue (left end )<br>Parameter2 quietValue (rest value)<br>Parameter3 maxValue (right end)                                                                                                                 |
|                      | Parameter value ranges:              | minValue < maxValue<br>minValue >= -60.0 Newton<br>maxValue <= 60.0 Newton                                                                                                                                               |
|                      | Command example:<br>System response: | <b>BarGraphValue -10 0 10</b><br>.Next position: 120.015 mm, Distance: 120.015 mm, Speed: 149.987 mm/s<CR><br>.Next position: 1890 step, Distance: 1890 step, Speed: 2362.000 steps/s<CR><br>.M_start<CR><br>.M_stop<CR> |

|                |                                      |                                                                                                                                                                                                                  |
|----------------|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>MovePos</b> | Command Description:                 | Movement to an absolute position                                                                                                                                                                                 |
|                | Syntax:                              | „MovePos <i>Parameter1 Parameter2</i> “                                                                                                                                                                          |
|                | Parameter units                      | Parameter1 in mm<br>Parameter2 in mm/s                                                                                                                                                                           |
|                | Parameter value ranges:              | 0.0 mm ≥ Parameter1 ≥ 120.014999 mm<br>0.0 > Parameter2 ≥ 444.50 mm/s                                                                                                                                            |
|                | Command example:<br>System response: | <b>MovePos 50 150</b><br>.Next position: 49.9745 mm, Distance: 49.9745 mm, Speed: 149.9870 mm/s<CR><br>.Next position: 787 step, Distance: 787 step, Speed: 2362.0000 steps/s<CR><br>.M_start<CR><br>.M_stop<CR> |

|                |                                      |                                                                                                                                                                                                                   |
|----------------|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>MoveRel</b> | Command Description:                 | Movement with a relative distance.                                                                                                                                                                                |
|                | Syntax:                              | „MoveRel <i>Parameter1 Parameter 2</i> “                                                                                                                                                                          |
|                | Parameter unit                       | Parameter1 in mm<br>Parameter2 in mm/s                                                                                                                                                                            |
|                | Parameter value range:               | - (current Position [mm]) ≥ Parameter1 ≥ (120.014999 mm - current Position [mm])<br>0.0 > Parameter2 ≥ 444.50 mm/s                                                                                                |
|                | Command example:<br>System response: | <b>MoveRel 50 390</b><br>.Next position: 99.9490 mm, Distance: 49.9745 mm, Speed: 389.9535 mm/s<CR><br>.Next position: 1574 step, Distance: 787 step, Speed: 6141.0000 steps/s<CR><br>.M_start<CR><br>.M_stop<CR> |

| O         | Command                                | Sets the format for displaying data from the force sensor. It allows choosing between not displaying the data (0) or displaying the data in grams (1) or visually as a bar graph (2).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |           |         |      |       |      |       |      |       |       |       |       |       |       |       |      |       |       |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |       |       |       |       |       |       |       |       |       |       |       |       |      |       |
|-----------|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|---------|------|-------|------|-------|------|-------|-------|-------|-------|-------|-------|-------|------|-------|-------|-------|------|-------|------|-------|------|-------|------|-------|------|-------|------|-------|------|-------|------|-------|------|-------|------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|-------|
|           | Description:                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |           |         |      |       |      |       |      |       |       |       |       |       |       |       |      |       |       |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |       |       |       |       |       |       |       |       |       |       |       |       |      |       |
|           | Syntax:                                | „O Parameter1“                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |           |         |      |       |      |       |      |       |       |       |       |       |       |       |      |       |       |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |       |       |       |       |       |       |       |       |       |       |       |       |      |       |
|           | Parameter value:                       | 0, 1 or 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |           |         |      |       |      |       |      |       |       |       |       |       |       |       |      |       |       |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |       |       |       |       |       |       |       |       |       |       |       |       |      |       |
|           | Command examples and system responses: | <div>O0</div> <div>O1</div> <div>.Load-cell: 0.4grams; 0.00 N&lt;CR&gt;<br/>.Load-cell: 201.8grams; 1.98 N&lt;CR&gt;<br/>.Load-cell: 272.2grams; 2.67 N&lt;CR&gt;<br/>.Load-cell: 0.4grams; 0.00 N&lt;CR&gt;</div> <div>O2</div> <div></div> <table><thead><tr><th>Force (N)</th><th>Command</th></tr></thead><tbody><tr><td>0.00</td><td>N&lt;CR&gt;</td></tr><tr><td>4.31</td><td>N&lt;CR&gt;</td></tr><tr><td>0.00</td><td>N&lt;CR&gt;</td></tr><tr><td>-0.02</td><td>N&lt;CR&gt;</td></tr><tr><td>-3.70</td><td>N&lt;CR&gt;</td></tr><tr><td>-1.12</td><td>N&lt;CR&gt;</td></tr><tr><td>0.00</td><td>N&lt;CR&gt;</td></tr><tr><td>-0.01</td><td>N&lt;CR&gt;</td></tr><tr><td>1.60</td><td>N&lt;CR&gt;</td></tr><tr><td>1.31</td><td>N&lt;CR&gt;</td></tr><tr><td>2.03</td><td>N&lt;CR&gt;</td></tr><tr><td>2.15</td><td>N&lt;CR&gt;</td></tr><tr><td>2.45</td><td>N&lt;CR&gt;</td></tr><tr><td>2.85</td><td>N&lt;CR&gt;</td></tr><tr><td>3.05</td><td>N&lt;CR&gt;</td></tr><tr><td>1.27</td><td>N&lt;CR&gt;</td></tr><tr><td>1.36</td><td>N&lt;CR&gt;</td></tr><tr><td>0.15</td><td>N&lt;CR&gt;</td></tr><tr><td>-0.58</td><td>N&lt;CR&gt;</td></tr><tr><td>-0.94</td><td>N&lt;CR&gt;</td></tr><tr><td>-1.00</td><td>N&lt;CR&gt;</td></tr><tr><td>-0.94</td><td>N&lt;CR&gt;</td></tr><tr><td>-0.56</td><td>N&lt;CR&gt;</td></tr><tr><td>-0.36</td><td>N&lt;CR&gt;</td></tr><tr><td>0.00</td><td>N&lt;CR&gt;</td></tr></tbody></table> | Force (N) | Command | 0.00 | N<CR> | 4.31 | N<CR> | 0.00 | N<CR> | -0.02 | N<CR> | -3.70 | N<CR> | -1.12 | N<CR> | 0.00 | N<CR> | -0.01 | N<CR> | 1.60 | N<CR> | 1.31 | N<CR> | 2.03 | N<CR> | 2.15 | N<CR> | 2.45 | N<CR> | 2.85 | N<CR> | 3.05 | N<CR> | 1.27 | N<CR> | 1.36 | N<CR> | 0.15 | N<CR> | -0.58 | N<CR> | -0.94 | N<CR> | -1.00 | N<CR> | -0.94 | N<CR> | -0.56 | N<CR> | -0.36 | N<CR> | 0.00 | N<CR> |
| Force (N) | Command                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |           |         |      |       |      |       |      |       |       |       |       |       |       |       |      |       |       |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |       |       |       |       |       |       |       |       |       |       |       |       |      |       |
| 0.00      | N<CR>                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |           |         |      |       |      |       |      |       |       |       |       |       |       |       |      |       |       |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |       |       |       |       |       |       |       |       |       |       |       |       |      |       |
| 4.31      | N<CR>                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |           |         |      |       |      |       |      |       |       |       |       |       |       |       |      |       |       |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |       |       |       |       |       |       |       |       |       |       |       |       |      |       |
| 0.00      | N<CR>                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |           |         |      |       |      |       |      |       |       |       |       |       |       |       |      |       |       |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |       |       |       |       |       |       |       |       |       |       |       |       |      |       |
| -0.02     | N<CR>                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |           |         |      |       |      |       |      |       |       |       |       |       |       |       |      |       |       |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |       |       |       |       |       |       |       |       |       |       |       |       |      |       |
| -3.70     | N<CR>                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |           |         |      |       |      |       |      |       |       |       |       |       |       |       |      |       |       |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |       |       |       |       |       |       |       |       |       |       |       |       |      |       |
| -1.12     | N<CR>                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |           |         |      |       |      |       |      |       |       |       |       |       |       |       |      |       |       |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |       |       |       |       |       |       |       |       |       |       |       |       |      |       |
| 0.00      | N<CR>                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |           |         |      |       |      |       |      |       |       |       |       |       |       |       |      |       |       |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |       |       |       |       |       |       |       |       |       |       |       |       |      |       |
| -0.01     | N<CR>                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |           |         |      |       |      |       |      |       |       |       |       |       |       |       |      |       |       |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |       |       |       |       |       |       |       |       |       |       |       |       |      |       |
| 1.60      | N<CR>                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |           |         |      |       |      |       |      |       |       |       |       |       |       |       |      |       |       |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |       |       |       |       |       |       |       |       |       |       |       |       |      |       |
| 1.31      | N<CR>                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |           |         |      |       |      |       |      |       |       |       |       |       |       |       |      |       |       |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |       |       |       |       |       |       |       |       |       |       |       |       |      |       |
| 2.03      | N<CR>                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |           |         |      |       |      |       |      |       |       |       |       |       |       |       |      |       |       |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |       |       |       |       |       |       |       |       |       |       |       |       |      |       |
| 2.15      | N<CR>                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |           |         |      |       |      |       |      |       |       |       |       |       |       |       |      |       |       |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |       |       |       |       |       |       |       |       |       |       |       |       |      |       |
| 2.45      | N<CR>                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |           |         |      |       |      |       |      |       |       |       |       |       |       |       |      |       |       |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |       |       |       |       |       |       |       |       |       |       |       |       |      |       |
| 2.85      | N<CR>                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |           |         |      |       |      |       |      |       |       |       |       |       |       |       |      |       |       |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |       |       |       |       |       |       |       |       |       |       |       |       |      |       |
| 3.05      | N<CR>                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |           |         |      |       |      |       |      |       |       |       |       |       |       |       |      |       |       |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |       |       |       |       |       |       |       |       |       |       |       |       |      |       |
| 1.27      | N<CR>                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |           |         |      |       |      |       |      |       |       |       |       |       |       |       |      |       |       |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |       |       |       |       |       |       |       |       |       |       |       |       |      |       |
| 1.36      | N<CR>                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |           |         |      |       |      |       |      |       |       |       |       |       |       |       |      |       |       |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |       |       |       |       |       |       |       |       |       |       |       |       |      |       |
| 0.15      | N<CR>                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |           |         |      |       |      |       |      |       |       |       |       |       |       |       |      |       |       |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |       |       |       |       |       |       |       |       |       |       |       |       |      |       |
| -0.58     | N<CR>                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |           |         |      |       |      |       |      |       |       |       |       |       |       |       |      |       |       |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |       |       |       |       |       |       |       |       |       |       |       |       |      |       |
| -0.94     | N<CR>                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |           |         |      |       |      |       |      |       |       |       |       |       |       |       |      |       |       |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |       |       |       |       |       |       |       |       |       |       |       |       |      |       |
| -1.00     | N<CR>                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |           |         |      |       |      |       |      |       |       |       |       |       |       |       |      |       |       |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |       |       |       |       |       |       |       |       |       |       |       |       |      |       |
| -0.94     | N<CR>                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |           |         |      |       |      |       |      |       |       |       |       |       |       |       |      |       |       |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |       |       |       |       |       |       |       |       |       |       |       |       |      |       |
| -0.56     | N<CR>                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |           |         |      |       |      |       |      |       |       |       |       |       |       |       |      |       |       |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |       |       |       |       |       |       |       |       |       |       |       |       |      |       |
| -0.36     | N<CR>                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |           |         |      |       |      |       |      |       |       |       |       |       |       |       |      |       |       |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |       |       |       |       |       |       |       |       |       |       |       |       |      |       |
| 0.00      | N<CR>                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |           |         |      |       |      |       |      |       |       |       |       |       |       |       |      |       |       |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |      |       |       |       |       |       |       |       |       |       |       |       |       |       |      |       |



|             |                                      |                                                                                                                                                                                                                                                                                          |
|-------------|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Q</b>    | Command                              |                                                                                                                                                                                                                                                                                          |
|             | Description:                         | Sets the interval for printing data cycles.                                                                                                                                                                                                                                              |
|             | Syntax:                              | „Q 10 <i>Parameter1</i> “                                                                                                                                                                                                                                                                |
|             | Parameter unit                       | Milliseconds                                                                                                                                                                                                                                                                             |
|             | Parameter value range:               | $50 \geq \text{Parameter1} \geq 50000$                                                                                                                                                                                                                                                   |
|             | Command example:                     | <b>Q 10 1000</b>                                                                                                                                                                                                                                                                         |
|             | System response:                     | <b>.Serial print interval: 1000&lt;CR&gt;</b>                                                                                                                                                                                                                                            |
| <b>t</b>    | Command                              | Resets the scale to zero                                                                                                                                                                                                                                                                 |
|             | Description:                         |                                                                                                                                                                                                                                                                                          |
|             | Syntax:                              | „t <i>Parameter1</i> “                                                                                                                                                                                                                                                                   |
|             | Parameter value:                     | 0 or 1                                                                                                                                                                                                                                                                                   |
|             | System responses:                    | <b>t 0</b><br><b>.Tare to Zero is complete&lt;CR&gt;</b>                                                                                                                                                                                                                                 |
|             |                                      | <b>t 1</b><br><b>Start known mass tare&lt;CR&gt;</b><br><b>Tare complete&lt;CR&gt;</b><br><b>Now, place your known mass on the loadcell.&lt;CR&gt;</b><br><b>Then send the weight of this mass in grams (i.e. m 100.0) from serial monitor.&lt;CR&gt;</b>                                |
| <b>m</b>    | Command                              | Setting the mass for the calibration procedure with known mass m. Used with the Tare command „t 1“.                                                                                                                                                                                      |
|             | Description:                         |                                                                                                                                                                                                                                                                                          |
|             | Syntax:                              | „m <i>Parameter1</i> “                                                                                                                                                                                                                                                                   |
|             | Parameter unit                       | grams                                                                                                                                                                                                                                                                                    |
|             | Command example and system response: | <b>m 50</b><br><b>50.000&lt;CR&gt;</b><br><b>.Known mass is: 50.00 grams&lt;CR&gt;</b><br><b>.New calibration value has been set to: 0.96, use this as calibration value (calFactor) in your project sketch.&lt;CR&gt;</b><br><b>.Save this value to EEPROM address 0? y/n&lt;CR&gt;</b> |
| <b>y</b>    | Command                              | Yes command.                                                                                                                                                                                                                                                                             |
|             | Description:                         |                                                                                                                                                                                                                                                                                          |
|             | Syntax:                              | „y“ (no arguments required)                                                                                                                                                                                                                                                              |
|             | System response:                     | <b>.Answer: yes&lt;CR&gt;</b><br>...                                                                                                                                                                                                                                                     |
| <b>n</b>    | Command                              | No command.                                                                                                                                                                                                                                                                              |
|             | Description:                         |                                                                                                                                                                                                                                                                                          |
|             | Syntax:                              | „n“ (no arguments required)                                                                                                                                                                                                                                                              |
|             | System response:                     | <b>.Answer: no&lt;CR&gt;</b><br>...                                                                                                                                                                                                                                                      |
| <b>Home</b> | Command                              | Moves to home position x = 0.00 mm at the last speed travelled                                                                                                                                                                                                                           |
|             | Description:                         |                                                                                                                                                                                                                                                                                          |
|             | Syntax:                              | „Home“ ( <i>No arguments required.</i> )                                                                                                                                                                                                                                                 |
|             | System response:                     | <b>.Moves the motor to the home position.&lt;CR&gt;</b><br><b>.M_start&lt;CR&gt;</b><br><b>.M_stop&lt;CR&gt;</b>                                                                                                                                                                         |



## Appendix: Logging file with execution of various commands, correct as well as with syntax errors.

```
##### <CR>
#START SETUP<CR>
#####<CR>
MTM_Setup V1.01<CR>
#####<CR>
.I2C_EEPROM_VERSION: 1.8.2<CR>
.EEPROM is connected.<CR>
<CR>
.The calibration value from the EEPROM is: 1115.49<CR>
.HX711 measured conversion time ms: 12.00<CR>
.HX711 measured sampling rate Hz: 83.33<CR>
.HX711 measured settlingtime ms: 194<CR>
.Load-Cell startup is complete<CR>
.Motor <DRIVER DM542T> initialization done!<CR>
#END SETUP<CR>
##### <CR>
.UNIT IS MILLIMETER<CR>
##### <CR>
S0
.ERR -1: <CR>
S 0
<CR>
.....<CR>
.Unit: mm<CR>
<CR>
.Max acceleration value [mm/s^2]: 635000.0<CR>
.Max acceleration value [steps/s^2]: 10000000.0<CR>
<CR>
.Max speed value [mm/s]: 444.50<CR>
.Max speed value [steps/s]: 7000.00<CR>
<CR>
.Min position [mm]: 0.0000<CR>
.Max position [mm]: 120.0150<CR>
.Min position [steps]: 0<CR>
.Max position [steps]: 1890<CR>
.....<CR>
L
.Current location of the motor [ mm]: 0.0000<CR>
.Current location of the motor [steps]: 0<CR>
Init
.M_start<CR>
.button_S3_pressed
.button_S3_released
.M_stop<CR>
MovePos 60 444
.Next position: 59.9440 mm, Distance: 59.9440 mm, Speed: 443.9920 mm/s<CR>
.Next position: 944 step, Distance: 944 step, Speed: 6992.0000 steps/s<CR>
.M_start<CR>
.M_stop<CR>
L
.Current location of the motor [ mm]: 59.9440<CR>
.Current location of the motor [steps]: 944<CR>
MovePos -60 444
.The new position is out of range. Min: 0 Max: 1890<CR>
.The new position is out of range. Min: 0.0000 Max: 120.0150<CR>
.Value is not within mm-range [0.0000 ... 120.0150]<CR>
MovePos 120 4444
.Value is not within speed-range [0.0000 ... 444.5000]<CR>
MovePos 120 444
.Next position: 119.9515 mm, Distance: 60.0075 mm, Speed: 443.9920 mm/s<CR>
.Next position: 1889 step, Distance: 945 step, Speed: 6992.0000 steps/s<CR>
.M_start<CR>
.M_stop<CR>
MoveRel -120 444
```



.Next position: 0.0000 mm, Distance: -119.9515 mm, Speed: 443.9920 mm/s<CR>

.Next position: 0 step, Distance: -1889 step, Speed: 6992.0000 steps/s<CR>

.M\_start<CR>

.M\_stop<CR>

L

.Current location of the motor [ mm]: 0.0000<CR>

.Current location of the motor [steps]: 0<CR>

MoveRel +120 444

.Next position: 119.9515 mm, Distance: 119.9515 mm, Speed: 443.9920 mm/s<CR>

.Next position: 1889 step, Distance: 1889 step, Speed: 6992.0000 steps/s<CR>

.M\_start<CR>

.M\_stop<CR>

L

.Current location of the motor [ mm]: 119.9515<CR>

.Current location of the motor [steps]: 1889<CR>

MoveRel -70 -666

.Value is not within speed-range [0.0000 ... 444.5000]<CR>

MoveRel -70 66

.Next position: 49.9745 mm, Distance: -69.9770 mm, Speed: 65.9765 mm/s<CR>

.Next position: 787 step, Distance: -1102 step, Speed: 1039.0000 steps/s<CR>

.M\_start<CR>

.M\_stop<CR>

O 1

.Load-cell: 0.1grams; 0.00 N<CR>

.Load-cell: -0.0grams; -0.00 N<CR>

.Load-cell: 0.0grams; 0.00 N<CR>

.Load-cell: 11.9grams; 0.12 N<CR>

.Load-cell: 454.4grams; 4.46 N<CR>

.Load-cell: 426.6grams; 4.18 N<CR>

.Load-cell: -0.1grams; -0.00 N<CR>

.Load-cell: -0.0grams; -0.00 N<CR>

.Load-cell: 0.0grams; 0.00 N<CR>

.Load-cell: -0.0grams; -0.00 N<CR>

O 2

-0.00 N <

-0.00 N <

-0.00 N <

3.21 N <

3.13 N <

4.66 N <

4.47 N <

5.50 N <

5.61 N <

6.19 N <

6.17 N <

4.57 N <

3.89 N <

2.36 N <

1.03 N <

0.19 N <

-0.48 N <

-1.11 N <

-1.17 N <

-1.38 N <

-1.36 N <

-1.70 N <

-0.00 N <

-0.00 N <

-0.00 N <

-0.00 N <

-0.00 N <

O 0

